Endometriosis AI Analysis Report

Advanced Multi-Modal Federated Prediction

1. Patient Clinical Metrics

Age	BMI	Pelvic Pain	Dysmenorrhea
41	31.2	9/10	8/10

Dyspareunia	Family Hist.	CA-125	Estradiol
Yes	No	95.0 U/mL	310.5 pg/mL

Progest.
3.0 ng/mL

2. AI Diagnostic Prognosis

Confidence Score: 40.8% (+/- 4.0%)

Estimated Stage: Stage II (Mild)

Future Risk Trajectory (Years 1, 3, 5):
1-Year: 48.3%, 3-Year: 46.3%, 5-Year: 53.8%

Generative Clinical Synthesis:

DISCLAIMER: AI research tool not a clinical diagnosis.

Primary Assessment: 40.8% (+/-4.0%) probability of active endometriosis. Estimated classification: Stage II (Mild).

Neural Routing: 47.9% of computation routed to the Superficial Peritoneal Expert Sub-Network.

Biomarker Highlights:
- CA-125: 95.0 U/mL elevated (threshold 35 U/mL). Differential diagnosis required.
- Estradiol: 310.5 pg/mL hyperestrogenism may drive ectopic proliferation.
- Pelvic Pain Score 9/10 indicates significant nociceptive involvement.

Recommendation: High-resolution transvaginal ultrasound and pelvic MRI for structural assessment.

Personal Actionable Plan:

Personal Actionable Health Plan (AI-generated for educational purposes only):

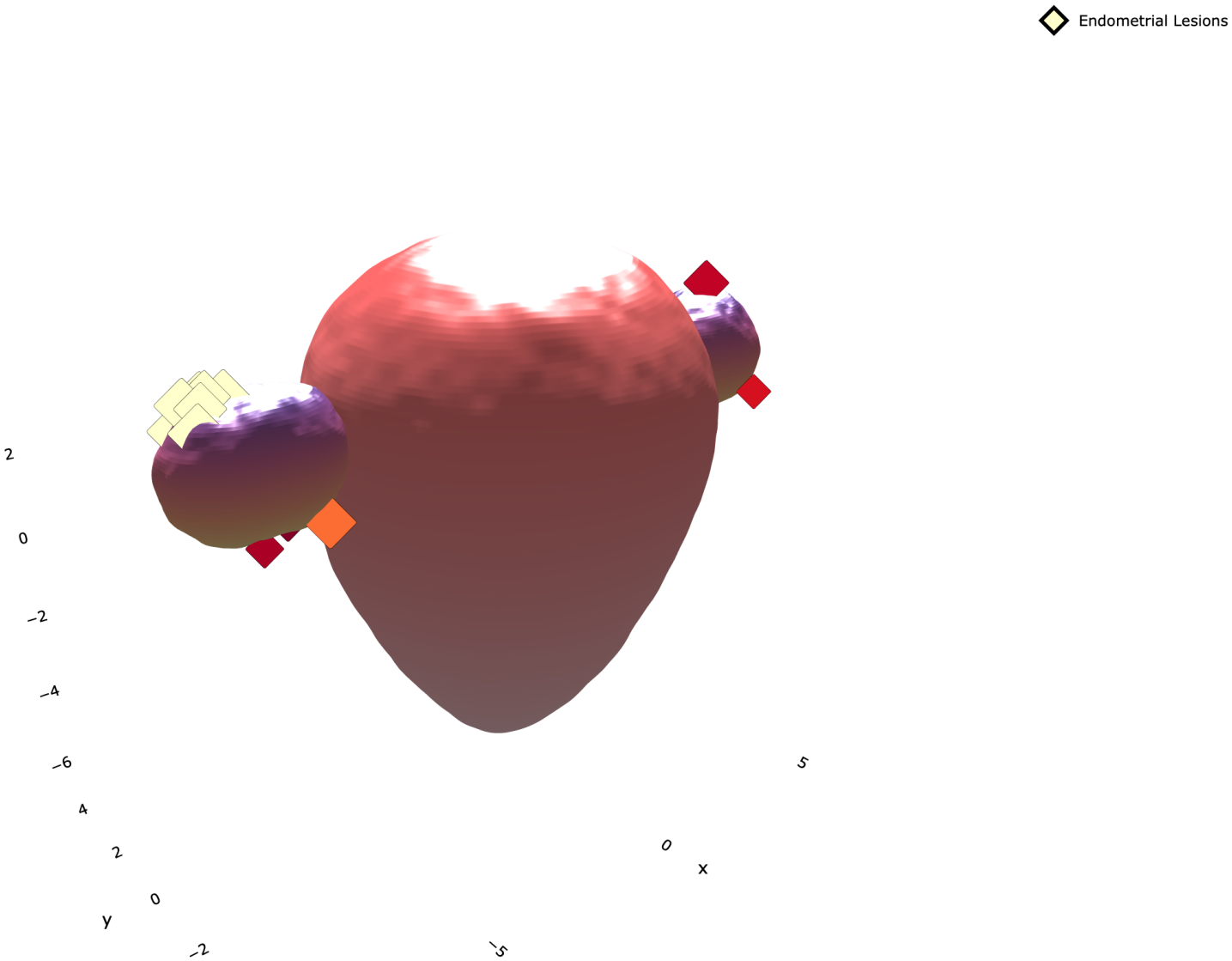
1. Monitoring: Risk profile is stable. Maintain routine gynaecological checkups.
2. Pain Management: Pain score is elevated discuss NSAIDs or pelvic floor physiotherapy.
3. Hormonal Balance: Omega-3-rich anti-inflammatory diet may help regulate excess oestrogen.
4. Lifestyle: An anti-inflammatory diet (Mediterranean) can reduce systemic inflammation markers.

Endometriosis AI Analysis Report

Advanced Multi-Modal Federated Prediction

3. Physics-Informed Digital Twin (3D Render)

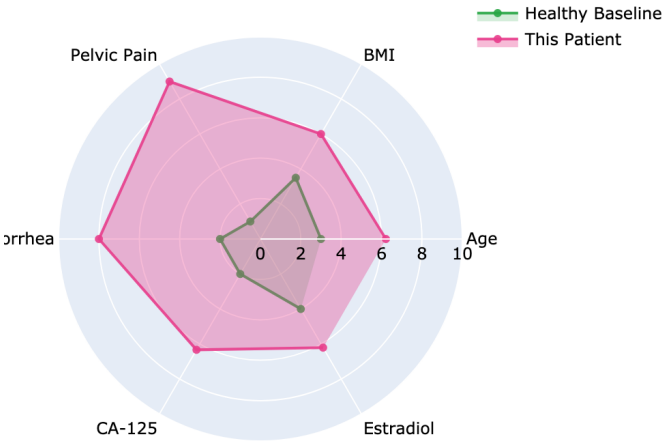
High-resolution rendering of the generated patient organ geometry, mapping lesions, endometriomas, and simulated adhesions based on the PINN predictions.

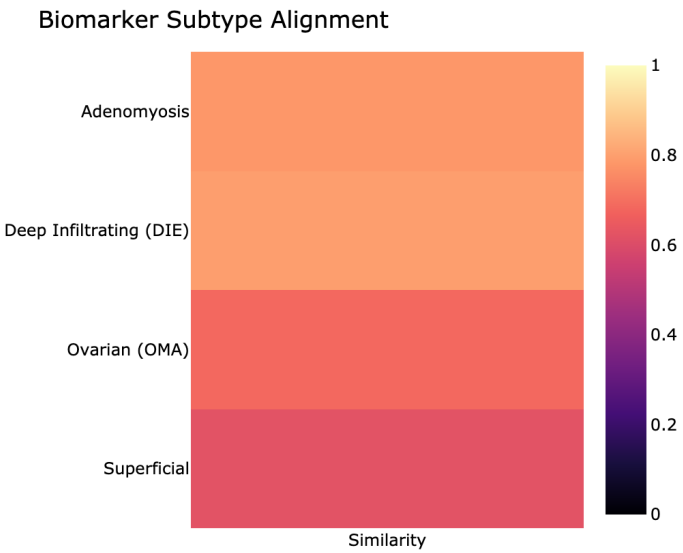


Endometriosis AI Analysis Report

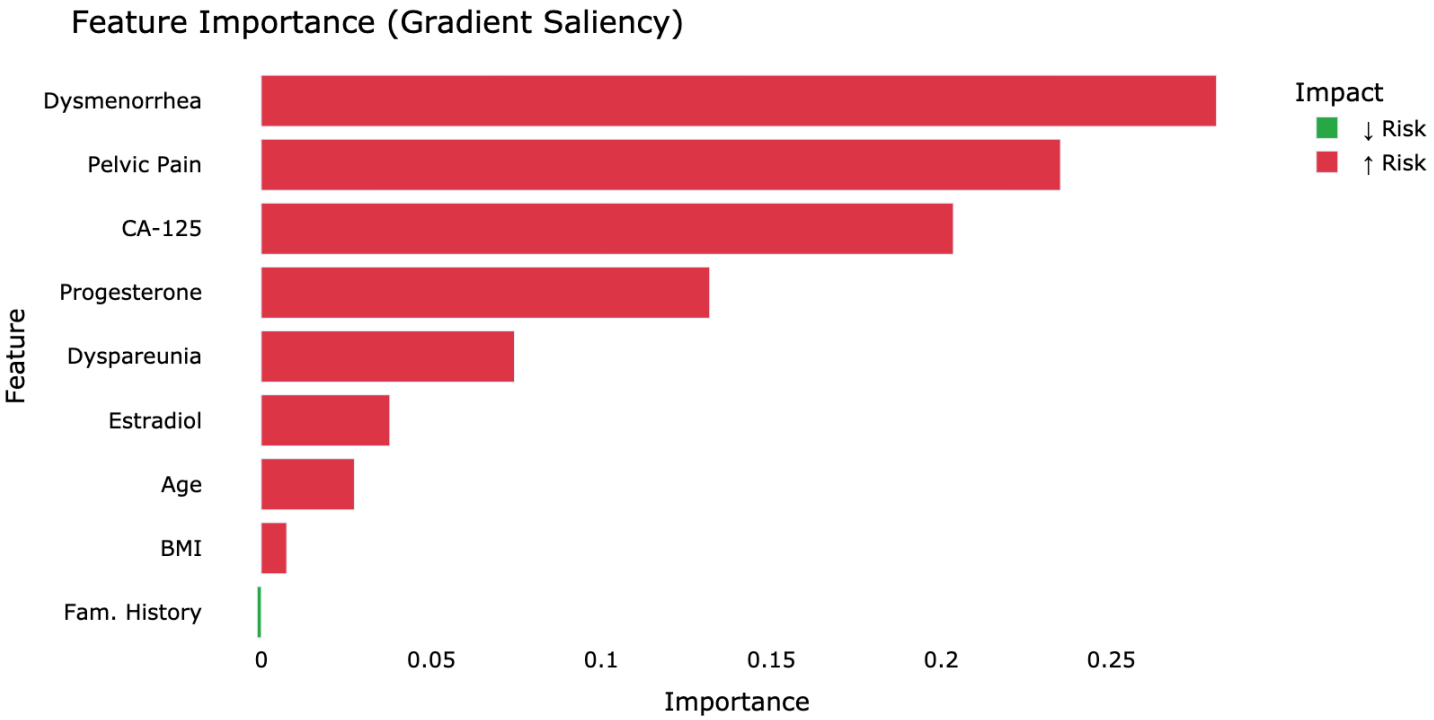
Advanced Multi-Modal Federated Prediction

4. Advanced Analytics & Explainable AI





Feature Importance (SHAP/XAI Explainer):



XAI Insight: Risk score 40.8%. Primary drivers: Pelvic Pain (9.0), Dysmenorrhea (8.0).